

QM CW2: Should a third runway be built at Heathrow Airport?

Introduction:

Heathrow airport, which opened in 1946, is of the greatest importance to the UK. This is shown by the 80 million passengers that it serves per year but also by its contribution to the UK's economy: More than 40% of the UK's exports go through Heathrow (Edgington, 2020), which has a very important impact on the UK's GDP. On the one hand, some would argue that, considering those existing numbers, but also with the world's economy expanding, and an estimation that demand for air transport will increase by an average of 4.3% per annum over the next 20 years (ICAO, 2020), the need to construct a third runway is fundamental. Nevertheless it shall not be forgotten the environmental costs of such a decision. This essay will analyze those by looking at the CO₂ emissions that Heathrow is responsible for. By using a fermi estimation approach, we will try to find a plausible answer to this research question: Would the construction of a third runway at Heathrow airport be rational, from an environmental point of view?

Literature review:

To answer our research question, This essay has tried to pick as objective and reputable sources as possible. The year of 2019, which was not affected by Covid, was chosen in order to lead an accurate analysis. The use of the "Heathrow travel report", Statista and the Climate Change Committee, provided us with factual information from 2019's airport statistics. The fermi-style approach of our essay required us, nevertheless, to formulate estimations of what would be the environmental consequences of such a construction, which remain hypotheses. We used the numbers estimated from "Airportwatch" which range between 700 000 and 740 000 to calculate the arithmetic mean to remain objective. But it shall be reminded to the reader that this data was found from a website which "aims to oppose any expansion of aviation and airports likely to damage the human or natural environment" (www.airportwatch.org.uk, 2023), and thus totally opposed to Heathrow's expansion. With this estimated data and factual statistics of the year 2019, we were able to estimate the total CO₂ emissions and the surplus resulting from this situation, as well as the increase of the share of Heathrow's involvement to total UK's aviation CO₂ emissions.

Methodology and results:

We will use a fermi-style approach to answer this question, we will assume that a third runway is constructed in order to make our estimations and to compare the impacts before and after the construction. As mentioned earlier, we will focus on CO₂ emissions.

Average CO₂ emissions by plane arriving or departing from Heathrow in 2019(2 runways):

We know that in the year 2019, the number of total planes that have departed from (237 921) and landed to(237 953)Heathrow was: $237\,921 + 237\,953 \approx 480\,000$ (Heathrow Travel Report, 2019)

We also know that total CO₂ emissions from Heathrow that same year were 16,2MillionsofTon(MT), (Statista Research Department, 2023).

Thus, average CO₂ emissions by plane in MT/CO₂ of that year were: $16,2/480=0,03375$

Heathrow's share in total emissions from UK aviation in 2019:

Total CO2 emissions from the UK's aviation in 2019 were 39.6MTCO2 in 2019(Climate Change Committee, 2020). Thus, Heathrow was responsible for: $16,2/39,6 * 100 \approx 41\%$ of these total CO2 emissions.

Heathrow's expected total CO2 emissions if 3rd runway:

To determine the number of flights arriving/departing to or from Heathrow if a third runway was built, we will use 700 000 as our lower bound and 740 000 as our upper bound (Clayton, 2016). The arithmetic mean is $(740\ 000 + 700\ 000)/2 = 720\ 000$ flights

We can now estimate the total CO2 emissions in MTCO2 that would be emitted from these 720 000 flights, which is: $0,03375*720=24,3$

Heathrow's share in total CO2 emissions if had a third runway been constructed with 2019 data:

Estimation of total CO2 emissions from UK aviation in MTCO2 with a third runway built at Heathrow in 2019: $39,6-16,2+24,3= 47,7$

Share of Heathrow's involvement in total CO2 emissions from UK's aviation in % under these assumptions: $(24,3/47,7)*100 \approx 51\%$

Heathrow's CO2 emissions report without and with a 3rd runway, based on 2019 reports	
number of flights arriving or departing from heathrow in 2019	480
Heathrow's CO2 emissions in MTCO2 2019 :	16,2
CO2 emissions by plane in MTCO2:	0,03375
UK's aviation total emissions in MTCO2 in 2019:	39,6
Share of CO2 emissions from Heathrow in 2019 in %:	40,9090909
estimated number of flights to arrive or depart if 3rd runway:	720
estimated total Heathrow CO2 emissions in 2019 if had been 3rd runway in MTCO3	24,3
Share of CO2 emissions from Heathrow if 3rd runway in %:	50,9433962

Conclusion:

This analysis shows that the construction of a third runway at Heathrow airport would seemingly lead to an environmental disaster as total CO2 emissions from Heathrow would be multiplied by 1,5: $16,2*1,5= 24,3$, or in other words, by 50%. In addition, this essay does not cover all environmental costs of this construction whereas there are multiple others: increase of poor air quality for the homes surrounding the airport, especially because of increase in car traffic (Mayor of London, 2016) as well as possible noise pollution. Nevertheless, this essay has not considered the possible improvements to public transports and construction of other Routes (Heathrow Travel Report, 2019) in case the runway is built, which could reduce the environmental costs. To conclude, this analysis presents many limits and should not be entirely reliable to provide a rational conclusion. Also, an economics approach to this question could lead to a totally opposed answer, supported by job-creation or GDP growth predictions.

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